

VELORA

An AI-Powered Dynamic Safe Speed Recommendation System
Technical Analysis Report

ADB AI for Safer Roads Innovation Challenge
SCMS School of Engineering and Technology , Karukutty
Center of Excellence SCMS Institute for Road Safety and Transportation (SiRST)

Project	Velora — Right Road. Right Time. Right Speed.
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Tech Stack	Python · Flask · scikit-learn · OSMnx · Leaflet.js · Pandas · SQLite
Challenge	Asian Development Bank × World Bank — AI for Safer Roads Innovation Challenge

1. Analytical Approach and Dataset Description

Velora adopts a **multi-layered, data-driven analytical approach** rooted in Safe System principles and Vision Zero. The platform fuses road-network geometry, infrastructure quality, behavioural exposure, crash history, weather, and real-time traffic into a unified risk model, producing a per-segment, temporally dynamic safe speed recommendation across all 5,498 mapped road segments in Bengaluru.

Dataset	Description / Processing
Road Network (OSMnx / GeoJSON)	Extracted via OSMnx from OpenStreetMap. Segments classified into National Highway (286), State Highway (380), and Urban Road (4,832). Stored as CSV and GeoJSON; geometry fields feed Leaflet map rendering and spatial joins.
Unified Platform (2.7 MB, 5,498 rows)	Master dataset synthesised from OSM geometry, GPS-probe 85th-percentile speed proxies, Mapillary-inspired infrastructure references (sidewalks, crosswalks, lighting), pedestrian and PTW exposure proxies, crash records, and land-use context. Statistical relationships from transport literature calibrate score distributions.
Traffic Temporal (~190 MB, 2.9 M rows)	Auto-generated by <code>dataset_bootstrap.py</code> at first run. Covers 6,066 segments × 480 timestamps (30-min intervals, 10-day window). Encodes congestion_score (0–1), category (Light / Moderate / Severe), and smoothed index using road class, time-of-day peaks, and random incident injection.

Dataset	Description / Processing
Crash Database (Seed CSV + SQLite)	Seeded from <code>crash_database_seed.csv</code> . Time-aware expiry: Minor crashes expire after 60 days; Major/Fatal after 80 days. Only active crashes feed into live risk re-scoring.
Hazard Events (SQLite)	Operator-reported events (roadworks, flooding, accidents) with date, time window, and optional manual speed override. Active hazards take absolute pipeline priority.
Weather (IoT Simulation)	Simulates Tipping Bucket Rain Gauge sensors (ESP32 + GSM/MQTT) refreshing every 30 min. Placement prioritises hotspots, flood corridors, and high-VRU zones. Monsoon-weighted distributions drive rainfall category (Clear → Extreme Storm), mapping to speed penalty (0 to −15 km/h).

1.1 Feature Engineering

Five composite scores feed the ML classifier directly:

Feature	Derivation
<code>road_function_score</code>	Road hierarchy complexity: intersection density, lane count, road class, and functional classification (arterial vs local).
<code>infrastructure_score</code>	Physical safety quality: sidewalks, pedestrian crossings, lighting quality, roadside hazard rating, median presence.
<code>exposure_score</code>	Road user vulnerability: pedestrian volume proxy, PTW share, school zone proximity, land-use type, urban/rural classification.
<code>human_tolerance_limit</code>	Biomechanical survivable crash speed (km/h) from Vision Zero literature. Represents the biological ceiling beyond which fatality probability rises sharply.
<code>operating_speed_score</code>	85th-percentile operating speed relative to posted limit — leading indicator of driver behaviour. Crash history excluded to avoid lagging-indicator bias.

Note: `crash_risk_score` is deliberately excluded from the ML feature set — it is a lagging indicator confounded by under-reporting. The classifier predicts *leading* structural risk independently of whether crashes have already occurred.

2. Model Architecture and Speed Safety Score

2.1 ML Classification Layer

The core classifier is a **Gradient Boosting** multi-class model (Random Forest as fallback) predicting whether a segment's posted speed limit is aligned with Safe System principles across four classes: **Aligned**, **Moderate Misalignment**, **High Misalignment**, and **Critical Misalignment**.

Both models were trained on the 5,498-row dataset with a stratified 80/20 split. Gradient Boosting (Accuracy 74.0%, F1 71.7%) is the default production model.

Metric	Random Forest	Gradient Boosting
Accuracy	72.6%	74.0%
Precision	69.7%	70.9%
Recall	72.6%	74.0%
F1-Score	70.8%	71.7%
CV Accuracy	75.8%	75.1%

2.2 Nine-Stage Safe Speed Pipeline

`prediction.py` runs nine sequential stages per segment per query. Later stages override earlier ones:

#	Stage	Logic
S1	ML Risk Score	GB predicts misalignment class and <code>ai_risk_probability</code> (0–100). $\text{road_risk_score} = 0.7 \times \text{ML_danger_mass} + 0.3 \times \text{crash_risk_score}$.
S2	Crash History Sync	$\text{crash_risk_score} = (\text{minor} \times 10 + \text{major} \times 25 + \text{fatal} \times 45) / \text{effective_length_km}$, capped at 100.
S3	Hotspot Scoring	$\text{hotspot_score} = 0.25 \times \text{misalignment} + 0.30 \times \text{exposure} + 0.35 \times \text{severity_density} + 0.10 \times (100 - \text{infrastructure})$. Classes: Safe / Moderate / High Risk / Severe Hotspot.
S4	AI Recommended Speed	$\text{ai_recommended_speed} = \min(\text{speed_p85}, \text{human_tolerance_limit})$. $\text{recommended_safe_speed} = \text{base} - \text{crash_risk} / \text{penalty}$ (NH:3, SH:4, Urban:6), clipped ≥ 20 km/h.
S5	Weather Adjustment	Clear=0, Light=−3, Moderate=−6, Heavy=−10, Extreme Storm=−15 km/h.
S6	Traffic Adjustment	Low=−0, Moderate=−7, Heavy=−15 km/h from temporal congestion score.
S7	School Zone Override	If <code>schools_count</code> >0, weekday, 08:00–16:00: $\text{speed} \times 0.6$, min 20 km/h.
S8	Hazard Event Override	Operator <code>temp_speed</code> bypasses all other stages (absolute priority).
S9	Road-Type Constraints	Urban Road: max 70, min 20 km/h. State Highway: min 25. National Highway: min 30 km/h. Hazard segments exempt from clamping.

2.3 Speed Safety Score Definition

`speed_safety_score` is a composite 0–100 index computed after Stages 1–2:

$$\text{speed_safety_score} = 100 - (\text{road_risk_score} \times 0.40 + \text{crash_risk_score} \times 0.30 + (100 - \text{infrastructure_score}) \times 0.30)$$

Component	Wt.	Rationale
road_risk_score	40%	ML-derived structural risk from road function, exposure, and operating speed.
crash_risk_score	30%	Severity-weighted crash density per km — lagging but objective.
Infrastructure Gap	30%	Penalises poor physical conditions: missing sidewalks, lighting, crosswalks.

Score 100 = structurally safe, crash-free, well-equipped segment. Score $\rightarrow$ 0 = Critical Misalignment blackspot requiring immediate intervention. Result is clipped to [0, 100] and rounded to one decimal place.

3. Validation Approach

3.1 ML Model Validation

Five complementary strategies were used to validate both classifiers:

- 1. Train-Test Split (80/20).** The 5,498-segment dataset was split into a stratified 80% training set (4,398 rows) and a held-out 20% test set (1,100 rows). Stratification preserved the class distribution across all four misalignment categories. Gradient Boosting achieved 74.0% accuracy and F1 71.7%, confirming generalisation beyond the training distribution.
- 2. K-Fold Cross-Validation (K=5).** Applied across five data partitions to assess model stability. Random Forest achieved mean CV accuracy of 75.8%; Gradient Boosting returned 75.1%. Consistency between test-set and cross-validated scores indicates low variance and confirms neither model is overfit to a specific subset of the road network.
- 3. Confusion Matrix Analysis.** Per-class precision, recall, and F1 were computed to identify class-level weaknesses. The classifier performs strongest on ‘Aligned’ and ‘High Misalignment’ segments, where structural signals are unambiguous. The highest confusion occurs between ‘Moderate Misalignment’ and ‘Aligned’ — an expected outcome given their overlapping feature distributions.
- 4. Feature Importance and SHAP.** Feature importances (RF: Gini impurity; GB: split gain) were computed and visualised. A SHAP summary plot is generated at runtime and surfaced in the platform’s XAI tab, enabling road safety engineers to audit per-segment risk drivers. `operating_speed_score` is the dominant predictor (~32–38% importance), confirming behavioural-speed misalignment as the strongest leading signal.
- 5. Pipeline Consistency Checks.** End-to-end tests confirmed the nine-stage pipeline produces monotonically safer speeds as risk inputs increase. School-zone and hazard-override modules were verified against time-boundary edge cases (overnight hazards, Sunday school exemption). Weather and traffic adjustment functions were tested at extreme inputs to confirm minimum-speed floor enforcement.

3.2 Dataset Representativeness

The unified dataset was synthesised using statistical relationships from peer-reviewed transport and biomechanics literature, ensuring realistic variable distributions. Operating speed proxies reflect GPS-probe 85th-percentile patterns typical of Bengaluru arterial and sub-arterial corridors. Infrastructure scores span the full range of Indian urban conditions — from well-maintained expressway service roads to narrow residential lanes without footways. Deliberate inclusion of outlier segments (e.g. high-speed corridors adjacent to schools) ensures the classifier is tested on the tail-risk cases most critical for road safety policy.

4. Limitations

4.1 Data Limitations

- **Synthetic operating speed data:** The 85th-percentile speed proxy is derived from statistical models, not direct GPS floating-car surveys. It may underrepresent speed variation on specific

corridors during special events or monsoon periods.

- **Simulated traffic data:** The 2.9-million-row temporal traffic dataset is probabilistic rather than from real-time loop-detectors or vehicle counts. Congestion at finer spatial and temporal scales (e.g. micro-junctions) will differ from simulation outputs.
- **IoT weather simulation:** The weather module mimics sensor behaviour without live hardware. Actual rainfall intensity, spatial variability, and flooding extent are not captured. Real deployment requires hardware commissioning and calibration.
- **Crash record completeness:** Seed data is subject to under-reporting — a documented challenge in India. Segments with few reported crashes may appear safer than they are.
- **Infrastructure scoring:** Scores are derived from Mapillary-inspired imagery references and OSM attributes, not formal on-site condition surveys. Scores may lag actual infrastructure changes.

4.2 Model Limitations

- **Classification accuracy:** At 74.0% accuracy, approximately one in four segments may be misclassified — most commonly between ‘Moderate Misalignment’ and ‘Aligned’. High-stakes intervention decisions should incorporate field validation.
- **Temporal stationarity:** The model was trained on a network snapshot. Structural changes (new construction, upgrades, lane reconfigurations) require dataset refresh and retraining.
- **Class imbalance:** ‘Critical Misalignment’ is rare in the Bengaluru network, potentially limiting recall for the highest-risk segments despite stratified sampling. SMOTE oversampling was not applied in the current version.
- **No causal inference:** The model identifies correlations, not causality. All speed-reduction associations are policy-informed, not experiment-derived.

4.3 Deployment and Scalability Limitations

- **Single-city training:** The model was trained on Bengaluru data only. Transfer to cities with substantially different road typologies may require fine-tuning or full retraining.
- **School zone detection:** `schools_count` relies on OSM point-of-interest completeness. Informal schools or newly established institutions may not be captured.
- **Real-time data integration:** Traffic and weather are currently simulated. Production deployment requires integration with live traffic management systems and physical IoT sensor networks.
- **Human-in-the-loop requirement:** The Hazard Management module depends on timely operator reporting. Automated hazard detection (computer vision, incident algorithms) is a recommended enhancement for operational deployment.

Despite these limitations, Velora demonstrates that a robust, explainable, and temporally dynamic safe speed recommendation system can be constructed from openly available geospatial data and machine learning — offering a scalable, low-cost pathway to Safe System implementation across ADB member cities.

*Built with Vision Zero principles for the ADB AI for Safer Roads Innovation Challenge
Focused on Bengaluru, India · Scalable to any city with OSM road network data*